

Practicalities at start of course: Finite Element Methods (41525)

Lectures	Wednesdays at 8am in room 074, Building 421. Duration: around 2 hours. Smaller follow-up lectures may be given during the day.
Computer exercises	003, 010, 014, 018 in building 421.
Teachers	Ole Sigmund (OS), Department of Civil and Mechanical Engineering, Solid Mechanics, Building 404, room 134, e-mail: olsi@dtu.dk Casper S. Andreasen (CSAN), Department of Civil and Mechanical Engineering, Solid Mechanics, Building 404, room 010, e-mail: csan@dtu.dk
Teaching Assistants	Victor Pissinger, email: vmapi@dtu.dk Jonathan T. Pedersen-Bjergaard, email: jtau@dtu.dk Brandon Parks, email: bmapa@dtu.dk Junpeng Wang, email: junwa@dtu.dk
Office hours	Office hours will be announced before the assignment periods. Often email requests are more efficient but please do not send same email separately to several TA's.
Apologies	Don't miss classes! In case of Force Majeure please send an e-mail to the teachers.
Literature	Cook, Malkus and Plesha, "Concepts and Applications of Finite Element Analysis", 4th edition, 2002, John Wiley and Sons. Buy it in the bookstore (Building 101) - see https://polyteknisk.dk/41525-finite-element-metoder-dtu . Handouts during the course.
Technical problems	If you have technical problems in the M-Data-bar please contact CONSTRUCT-support (icon on the desktop).
Homework	Course notes have one chapter pr. course week. Each chapter also refers to recommended reading in the Cook book. Further info and material may be found on DTU Learn.
Assignments and examination	The course evaluation consists of the evaluation of three individualized group reports and a grade can only be awarded if the individual student's contribution to the project can be ascertained. It must be clearly specified for which sections each student has the (main) responsibility. A group project is not deemed to be individualized if the students merely state that they have contributed equally to all sections of the report or the like. Each group will consist of maximum two students. The reports (respectively 4, 5 and a maximum of 15 pages) must demonstrate and discuss solutions of the assignments. The final grade is based on the overall performance the individual student has demonstrated in the three reports.

AI Use

DTU embraces use of AI. However, its use must be declared in accordance with DTU and course guidelines. See a.o. DTU Code of honour:

<https://student.dtu.dk/en/exam/exam-cheating/dtu-code-of-honour>
and

<https://student.dtu.dk/en/exam/exam-cheating>.

While AI tools may be used to support tasks such as coding, data processing, or drafting text, students remain fully responsible for all aspects of the submitted work and must be able to explain, justify, defend and explain the codes, methods, analyses, and conclusions presented in the report.

As part of the feedback we provide on the reports, we will also assess whether the declared use of AI appears consistent with the submitted work. If possible inconsistencies are identified, we will invite groups to a discussion clarifying the matters and, where relevant, address potential violations of DTU's plagiarism and code of honor policies.

- Assignments**
- 1) Full assignment published on DTU Learn 23/9 (week 4), 1pm. To be uploaded to DTU Learn by 29/9 at 10pm (maximum 4 pages + cover page).
 - 2) Full assignment published on DTU Learn 4/11 (week 9), 1pm. To be uploaded to DTU Learn by 10/11 at 10pm (maximum 5 pages + cover page).
 - 3) Full assignment published in class 11/11 (week 10). To be uploaded to DTU Learn by 7/12 at 1pm (maximum 15 pages + cover page).

Tentative course schedule

Week	Date	Lecturer	Topic
1	2/9	OS	Matlab truss FE-analysis
2	9/9	CSAN	Matlab truss FE-analysis (material non-linearity)
3	16/9	BNL/CSAN	Matlab truss FE-analysis (geometrical non-linearity)
4	23/9	OS	Matlab truss FE-analysis (Topology Optimization)
			Publishing of first full report assignment on DTU Learn (1pm)
	29/9		First report (upload no later than 10pm)
5	30/9	OS	Continuum type FE-analysis
6	7/10	CSAN	Continuum type FE-analysis (Equation solving/stress computations/distributed forces)
		CSAN	Feedback to first report
	14/10		FALL BREAK
7	21/10	CSAN	Continuum type FE-analysis (energy principles).
8	28/10	OS	Continuum type FE-analysis (Isoparametric formulation)
9	4/11	CSAN	Dynamics (eigenvalue analysis)
			Introduction of final projects
			Publishing of second full report assignment on DTU Learn (1pm)
	10/11		Second report (upload no later than 10pm)
10	11/11	CSAN	Alternative methods and High Performance Computing. Final assignment.
11	18/11	OS	Topology Optimization. Final assignment.
		OS	Feedback to second report
12	25/11	OS	Error estimates + Mesh-generation. Final assignment.
13	2/12	CSAN/OS	Course evaluation. Final assignment.
	7/12		Hand in third report on final assignment before 1pm

Final projects (may be subject to change!)

Project	Description
1	Plate modelling
2	Geometrical non-linearity
3	Transient problems
4	Plasticity
5	Incompressibility/Stokes flow
6	Special elements
7	Your own idea ...

Good luck with the course!

Casper S. Andreasen and Ole Sigmund